Monday, January 05, 2026 3:49 PM

- Intro to HTML → it is not a programming language.
  - ↳ it is a way to define the formatting and structure the content of a webpage.
  - It does not contain
    - logic
    - use loops or conditional statement

Boilerplate of HTML

```
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

</body>
</html>
```

→ tells the browser that the page is using HTML5.

Programming.
- Python
- Java
- C++.

Markup language
- XML
- SGML.
- HTML.

- Two type of Tags.
  - Tags with closing tag.
  - Self closing tag.

- Self closing tag.
  - they do not require a self closing tag.
  - they are complete by themselves.

Characteristics
- They do not wrap the content
- They do not closing counterparts.
- They are used mainly for
  - meta data <meta> — provide metadata information to the browser
  - line break <br>. —
  - Image <image> —
  - Inputs <input>. —

2. Tags that require opening and closing tag.
   - These tags wrap the content and therefore must have both opening and closing tag.

Characteristics
- They contain text or other tags inside them
- They always come in pair

Eg. <head> .... </head>
<body> ... </body>
<html> ... </html>
 .... .

- Purpose of a markup language.

→ To structure and format content.

Whatever content you have:

- text
- image
- video
- forms
- Links

We use HTML to structure how the content appears on a webpage

HTML Structure

HTML

- Head
  - Title. – page title shown in browser tab
  - meta information
  - CSS styling.
  - Java script

  Nothing visible on the webpage appear from <head>.

- Body.

  The body contains everything the user see:
  - Heading.
  - Paragraph. ... .
  - Images. <img>.
  - Videos.
  - Forms
  - Buttons.
  - Links
  - menu.
  - ads.
  - Content of webpage

  If it is visible on the screen it belongs to <body> </body>.

<h1> Heading </h1>. → Largest head

<h2> </h2>

<h3> </h3>

⋮

<h6> </h6> → smallest.

- Hyperlink (Anchor tag)

  If you want to put a URL or a hyperlink on a webpage, we have to use anchor tag.

  <a>.

- It uses an attribute called href
  href contain the URL.

Eg.

<a href = 'URL' > Click here </a>.

→ visible text on the webpage.

↓ destination URL

- Multimedia in HTML

- Video Tag.

```
<video controls>
    <source src = 'video.mp4' type = 'video/mp4'>
</video>.
```

- Audio tag.

```
<audio controls>
    <source src = 'audio.mp3' type = 'audio/mpeg'>.
</audio>.
```

Ordered & unorder list

- ordered list (<ol>), → used when order matters.

```
<ol>.
    <li> Python </li>
    <li> Data Analysis </li>
    <li> Machine learning </li>
    <li> Deep Learning </li>
</ol>.
```

→

1. Python
2. Data Analysis
3. Machine Learning
4. Deep Learning

- unordered list <ul>.

```
<ul>
    <li> Python </li>
    ⋮
</ul>
```

- Python
- Data Analysis
- Machine Learning.

<li> → list items.
every item inside ol & ul must be wrapped inside <li>.

- Input tag. → self closing tag., it uses some attribute to define behaviour.

<input type = 'text' >

- Characters appear as dots or asterisk.
- used for security.
- [illegible] hides actual input.

- Browsers make default input

HTML supports many input types
- text
- number
- email
- password.
- checkbox
- radio
- search
- submit

HTML → structure & layout.

CSS → beautification of content.
Cascading style sheet

In order to send data from client → server we will use HTML input components.

Type of HTML input components
- Text box
- Number box
- Password box
- Search box
- Rate button
- Checkboxes
- Submit box

A good application should.
- Ask the user for input via a UI.
- Let the client / user click a button
- Send data automatically to the backend.
- Show the result clearly

- Understanding input tag.

  <input type = 'text'>.
- It is a self closing tag.

The role of <form> Tag.

Data can never be sent from frontend to backend without a <form> tag.

- what it does.
  - collect the data from all input fields inside it
  - sends the data back to the backend.
  - sends data only when the submit button is clicked.

without <form>.
- Input are just UI elements
- They do not communicate with the server.

Two critical attributes of the <form> tag.

These two tags decides where & how data is sent

- action → define which backend route should recive the data
- method. → define how data is sent
  - Usually
    - GET
    - POST.

- why name attribute is mandatory

without name, backend cannot identify input values

<input type = 'text' name = 'a' > → frontend.

Backend recive

request.argv.get('a')

No name → no data.